

AI-POWERED NETWORK INTRUSION DETECTION SYSTEM (NIDS)

WEEK 1 & WEEK 2 PROGRESS PRESENTATION(MILESTONE-I)

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PROJECT INTRODUCTION

- ❑ Network security is critical due to increasing cyber attacks.
- ❑ Traditional security systems depend on rule-based detection.
- ❑ These systems fail to detect **unknown or new attacks**.
- ❑ Machine Learning helps detect hidden patterns in network traffic.
- ❑ **Objective:** Build an **AI-based intrusion detection system**.

GOALS OF THE PROJECT

- ❑ Detect malicious network traffic using AI.
- ❑ Classify traffic as **Normal** or **Attack**.
- ❑ Analyze patterns in network traffic data.
- ❑ Improve detection accuracy and reduce false alarms.
- ❑ Generate alerts for suspicious activity.

NETWORKING FUNDAMENTALS

- **TCP (Transmission Control Protocol)**
 - Connection-oriented
 - Reliable communication
 - Uses 3-way handshake
- **UDP (User Datagram Protocol)**
 - Connectionless
 - Faster transmission
 - Used in real-time applications
- Other protocols: **HTTP, HTTPS, DNS, SSH, ICMP**

TYPES OF NETWORK ATTACKS

- Common attacks analyzed in the dataset:
- DDoS (Distributed Denial of Service)
- Brute Force Attacks
- Port Scanning
- Injection Attacks
- SYN Flood
- Observation: **Most attacks target TCP-based services.**

DATASET ACQUISITION

- Datasets used:
- **NSL-KDD Dataset**
- **CICIDS2017 Dataset**
- These datasets contain labeled network traffic including **normal traffic and multiple attack types**.

DATASET FEATURES

- Important features studied:
- duration
- protocol_type
- service
- flag
- src_bytes
- dst_bytes
- class label
- Purpose: Understand **traffic behavior and attack patterns**.

DATASET ANALYSIS

- Analysis performed:
- Distribution of **normal vs attack traffic**
- Relationship between **protocol and attack type**
- Feature correlation
- Identification of **class imbalance**

MULTI-DIMENSIONAL DATA ANALYSIS

- Techniques used:
- **Slicing** – Selecting data based on one condition
- **Dicing** – Applying multiple filters
- **Roll-up** – Aggregating data
- **Drill-down** – Viewing detailed data
- **Pivoting** – Changing data perspective

DATA CLEANING

- Steps performed:
- Removed duplicate records
- Checked missing values
- Removed irrelevant features
- Purpose: Improve **data quality and model performance**.

DATA PREPROCESSING

- Feature engineering techniques:
- Encoding categorical features
- Log transformation for skewed features ($\log(1+x)$)
- Normalization of numerical features
- Dataset split into **training and testing sets**.

MACHINE LEARNING WORKFLOW

- Pipeline followed:
- Data Extraction
- Data Cleaning
- Feature Engineering
- Model Training
- Model Evaluation
- Prediction & Alert Generation
- Evaluation metrics include **Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.**